English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Applied Mathematics / Mathematics, 11 June 2025 Examination – *To pass this examination, demonstrate at least one of the following: familiarity with what was required in previous assessments; knowledge of the material covered in tutorials; or mastery of the material and the ability to apply it in a new context. The grade equals the number of points; 13 points are available. Time allowed: 4 hours. Good luck!*

- *Problem 1 contains questions from this year's and last year's tests.*
- *Problem 2 contains questions from the list provided at the start of the semester.*
- *Problem 3 contains new questions.*

Problem 1. 1. Let n be a positive integer. Choose u_1 uniformly from $\{1, \dots, n\}$, then u_2 uniformly from $\{1, \dots, u_1 - 1\}$, and so on until $u_\ell = 1$. Set $E_n = \{u_1, \dots, u_\ell\}$.

(a) (0.25 points) Find $\mathbb{P}(k \in E_n)$ for $k \leq n$.

(b) (0.25 points) Find $\mathbb{P}(2 \in E_n \mid 3 \notin E_n)$.

(c) (0.5 points) Find $\mathbb{E}[|E_n|]$ and $D(|E_n|)$ and estimate their asymptotic behaviour as $n \to \infty$.

2. (0.5 points) Each face of a six-sided die is filled with a uniformly random number from $\{1, \dots, 6\}$.

(a) What is the probability of rolling a six, and the expected number of rolls until this happens?

(b) What is the expected number of rolls until the product of all outcomes is first divisible by 6?

(c) If two rolls produced two sixes but the labels are unknown, what is the probability that all faces are sixes? Given the same observation, what is the most likely labelling?

3. Asen and five friends play football every day, choosing uniformly at random two teams of three. Find the expectation and variance of the number of days until:

(a) (0.25 points) Asen is first on a team with his two best friends;

(b) (0.25 points) a previously used division into teams first recurs;

(c) (0.5 points) every possible division has occurred.

4. (0.25 points) Let X be a random number in $\{0, 1, 2\}$ and $Y \sim \mathrm{Ber}(2/3)$ independent of it. Find the joint distribution and correlation of X and $Z = (X + Y) \pmod 3$.

5. (0.25 points) Let X be uniform on $\{1, \dots, 6\}$. Find its probability generating function and express $\mathbb{E}[X]$ and DX in terms of it.

6. (0.5 points) For nonnegative integer-valued X, Y, does independence imply $g_X(s)g_Y(s) = g_{X+Y}(s)$ for all $s \geq 0$? Does the converse hold? Prove your claims.

7. (0.5 points) Let X, Y be the results of two independent standard dice. Find the distribution of $X + Y$. Can two blank six-sided dice be labelled with other positive integers, not necessarily identically, to give the same sum distribution?

8. (0.5 points) Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, with a sample of n observations. Test $H_0 : \mu = 3.8$ against $H_1 : \mu \neq 3.8$ at level $\alpha = 0.05$. Can H_0 be rejected for $n = 16$, $\overline{X} = 3.295$, $S^2 = 1.4641$? Derive the test using likelihood ratios.

9. (0.5 points) Let $X \sim U(-\alpha, \alpha)$. Find method of moments and maximum likelihood estimators of α, and determine whether they are unbiased and/or consistent.

Problem 2. 1. (0.5 points) A coin with probability p of heads is tossed n times. Let E be the event that the first toss is heads and F_k the event of exactly k tails. For which integer pairs (n, k) are E, F_k independent?

2. (1 point) Let $(X_1, \dots, X_n)$ be a random permutation of $\{1, \dots, n\}$ and $S = X_1 + \dots + X_n$.

(a) Find $\mathbb{E}S$ and DS.

(b) Prove $D(X + Y) = DX + DY + 2\mathrm{Cov}(X, Y)$ for two random variables.

(c) Express $\mathbb{E}S$ using $\mathbb{E}X_i$. Find $\mathbb{E}X_i$, $\mathbb{E}X_i^2$ and DX_i for every i.

(d) Express DS using DX_i and $\mathrm{Cov}(X_i, X_j)$. Find the covariance for every i, j.

(e) (0.25 points) Let $U_1, U_2 \sim U(0, 1)$ be independent and $Y = \ln U_1 / \ln U_2$. Find the density and expectation of Y.

(f) (0.25 points) Is $1 + \lfloor \ln U_1 / \ln U_2 \rfloor$ geometrically distributed? Explain.

3. (1 point) Consider the standard distributions $\mathrm{Ber}(p)$, $\mathrm{Bin}(n,p)$, $\mathrm{Ge}(p)$ and $\mathrm{Poi}(\lambda)$. Let X_1, X_2 be iid with one of these laws. Is $X_1 X_2$, or $X_1 + X_2$, of the same distributional type as X_1, possibly with different parameters? Give a full justification in each of the four cases. If the answer is not always yes, can you give an example where this situation does occur?

4. (1 point) Find to accuracy 0.01 the probability that the first nonzero decimal digit of X is 1 when:

   (a) $X \sim N(0,1)$;

   (b) $X \sim \mathrm{Exp}(1)$;

   (c) $X = U_1/U_2$ with independent $U_1, U_2 \sim U(0,1)$. Also find $\mathbb{E}X$.

5. (0.5 points) Let $X, X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ be iid. Find maximum likelihood estimators of μ, σ, both unknown. Find a 95% confidence interval for μ when σ is unknown.

Problem 3. 1. Let $X, Y \sim \mathrm{Exp}(1)$ be independent, $S = X + Y$ and $W = X/(X+Y)$.

   (a) (0.75 points) Show that S, W are independent and find their distributions.

   (b) (0.25 points) Find $\mathbb{E}[W^k]$ for $k > -1$.

2. (0.5 points) Let $X, Y \sim \mathcal{N}(0,1)$ be independent and $Q = X/Y$. Find the distribution and density of Q.

3. Choose (X, Y) uniformly in the unit disc $x^2 + y^2 \le 1$.

   (a) (0.5 points) Find the density of its distance R from the centre.

   (b) (0.5 points) Find $\mathbb{E}R$, the median and the mode of R.

4. Let $T_1, \ldots, T_n$ be iid $\mathrm{Exp}(\lambda)$ with unknown $\lambda > 0$.

   (a) (0.25 points) Find the maximum likelihood estimator of λ.

   (b) (1 point) Prove $2\lambda \sum_{i=1}^n T_i \sim \chi^2_{2n}$ and derive an exact two-sided 95% confidence interval for λ.

   (c) (0.25 points) Use likelihood ratios to formulate a level 0.05 test of $H_0 : \lambda = 1$ against $H_1 : \lambda < 1$, and express the rejection region in terms of $\sum T_i$.